Ch 8.1: Decision Trees

Lecture 24 - CMSE 381

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Mon, Nov 4, 2024

Announcements

Last time:

- Cubic Splines

This lecture:

- 8.1 Decision Trees

Announcements:

- HW #7 Sun, 11/10
- Projects

Lec #	Date			Reading	HW
21	Mon	10/28	Polynomial & Step Functions	7.1,7.2	
22	Wed	10/30	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
23	Fri	11/1	Regression Splines	7.4	HW #6 Due Sun 11/3
24	Mon	11/4	Decision Trees	8.1	
25	Wed	11/6	Class Cancelled (Dr Munch out of town)		
26	Fri	11/8	Random Forests	8.2.1, 8.2.2	HW #7 Due Sun 11/10
27	Mon	11/11	Maximal Margin Classifier	9.1	
28	Wed	11/13	SVC	9.2	
29	Fri	11/15	SVM	9.3, 9.4	HW #8 Due Sun 11/17
30	Mon	11/18	Single layer NN	10.1	
31	Wed	11/20	Multi Layer NN	10.2	
32	Fri	11/22	CNN	10.3	HW #11 Due Sun 11/24
33	Mon	11/25	TBD: Unsupervised learning/clustering	12.1, 12.4?	
	Wed	11/27	Virtual: Project office hours		
	Fri	11/29	No class - Thanksgiving		
	Mon	12/2	Review		
	Wed	12/4	**Midterm #3**		
	Fri	12/6	No class - EGR Design Day		Project due

Section 1

Decision Trees

Big idea

- Previously, we've always had some form of function for prediction: $f(X_1, \cdots, X_p) = \hat{Y}$
- Now, we make a sequence of decisions to make a prediction, either regression or classification version

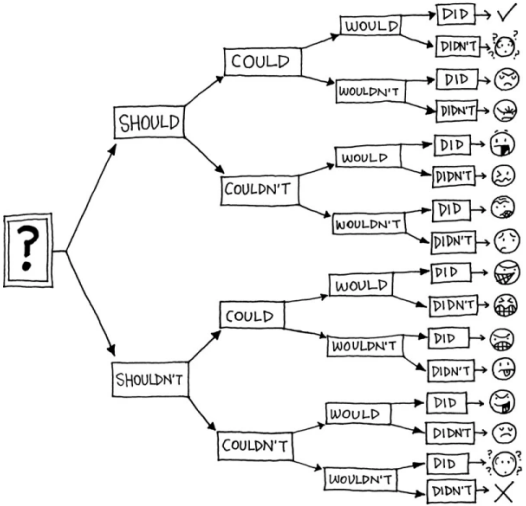

Image: https://marekbennett.com/2014/02/14/decision-tree/

Subset of Hitters data

	Hits	Years	Salary	LogSalary
1	81	14	475.0	6.163315
2	130	3	480.0	6.173786
3	141	11	500.0	6.214608
4	87	2	91.5	4.516339
5	169	11	750.0	6.620073
...	...	...	...	...
317	127	5	700.0	6.551080
318	136	12	875.0	6.774224
319	126	6	385.0	5.953243
320	144	8	960.0	6.866933
321	170	11	1000.0	6.907755

- Remove observations missing salary values
- log transform salary for something closer to bell shape
- Goal: predict log salary (can reverse by returning $\exp(x)$ if model returns x)

First decision tree example

	Hits	Years	LogSalary
1	81	14	6.163315
2	130	3	6.173786
3	141	11	6.214608
4	87	2	4.516339
5	169	11	6.620073
...	...	...	...
317	127	5	6.551080
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- Top split assigns observations with *Years* < 4.5 to left branch
- Return mean response for players with that.
- Predictions:
  - mean log salary is 5.107, so returns $\exp(5.107) = \$165.174$ thousand dollars
  - $5.999 \Rightarrow \$402,834$
  - $6.740 \Rightarrow \$845,346$

Interpretation of example

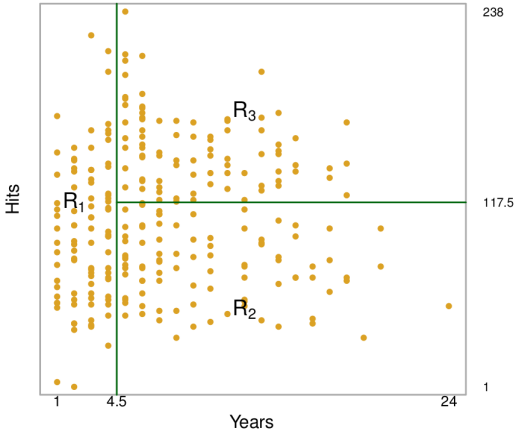
- Years most important factor for determining salary
- Players with less experience earn lower salaries than more experienced
- For the less experienced players, number of hits plays little role in salary
- For more experienced players, number of hits affects it
- Likely an oversimplification of real relationship, but easier to interpret and has nice graphical representation

Coding a regression decision tree

Regions defined by the tree

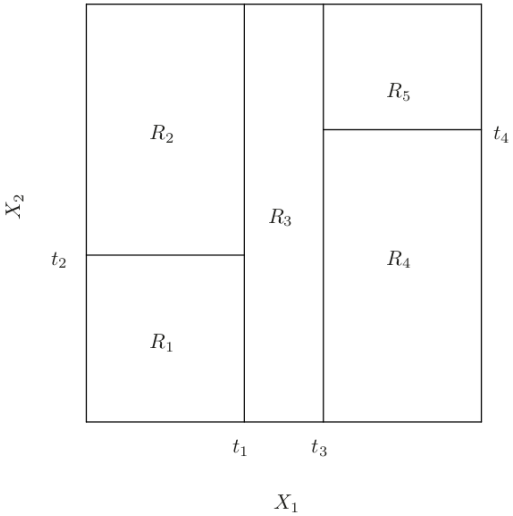
- Point out tree is "upside down"
- Leaves of the tree:
  - $R_1 = \{X \mid Years < 4.5\}$
  - $R_2 = \{X \mid Years >= 4.5, Hits < 117.5\}$
  - $R_3 = \{X \mid Years >= 4.5, Hits >= 117.5\}$
- Other splits are called Internal Nodes
- Segments that connect nodes called Branches or Edges

Viewing Regions Defined by Tree



How do we actually get the tree? Two steps

1. We divide the predictor space – that is, the set of possible values for $X_1, X_2, \cdots, X_p$ — into J distinct and non-overlapping regions, $R_1, R_2, \cdots, R_J$.
2. For every observation that falls into the region R_j, we make the same prediction = the mean of the response values for the training observations in R_j.



Step 1: How do we decide on R_js?

Goal:
Find boxes $R_1, \cdots, R_J$ that minimize

$$\sum_{j=1}^{J} \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2$$

$\hat{y}_{R_j}$ = mean response for training observations in jth box

- Can't actually check every possible partition
- Instead, go for top-down greedy approach called *Recursive binary splitting*
- Begins at top of tree with all data points
- Uses best split at every step

Recursive Binary Splitting

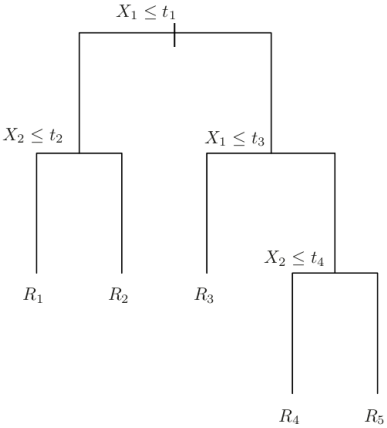
- Pick X_j
- Pick s so that splitting into $\{X \mid X_j < s\}$ and $\{X \mid X_j \geq s\}$ results in largest possible reduction in RSS

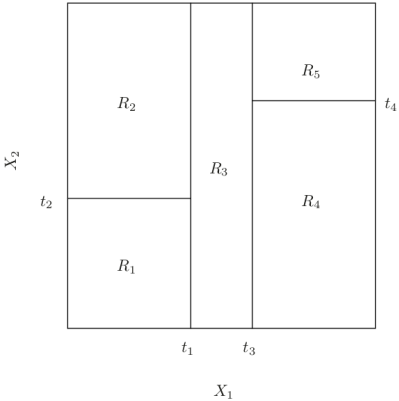
$$R_1(j,s) = \{X \mid X_j < s\}$$
$$R_2(j,s) = \{X \mid X_j \geq s\}$$

$$\sum_{i|x_i \in R_1(j,s)} (y_i - \hat{y}_{R_1})^2 + \sum_{i|x_i \in R_2(j,s)} (y_i - \hat{y}_{R_2})^2$$

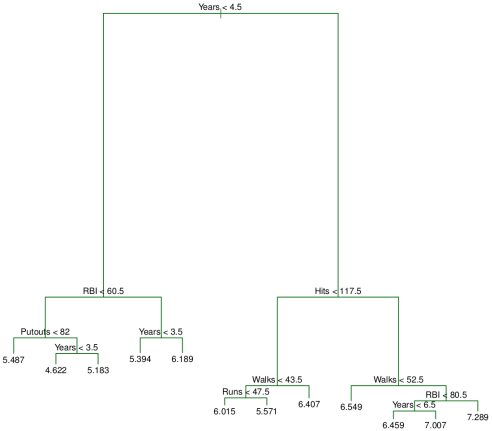
Rinse and repeat

- Do this repeatedly
- Each time can split one of the previously identified regions
- Keep going until some stopping criterion is reached. E.g. until each region has at most 5 observations



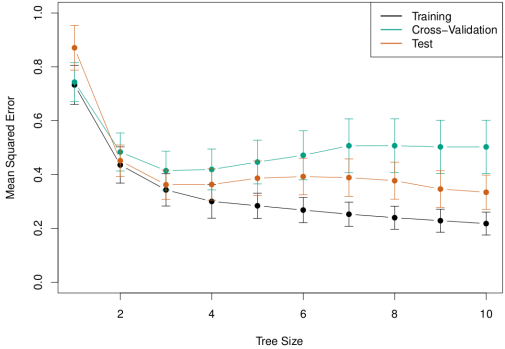


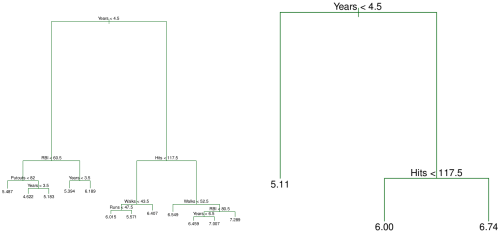
Pruning



- Big trees leave you open to potential overfitting
- Could just stop building earlier, but that's short sighted
- Instead, grow a big tree and prune it back
- Find subtree with best test error rate
- Too many subtrees to test them all

Result of pruning



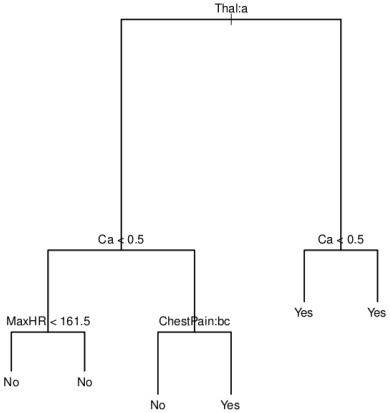


Result of pruning is the three leaf tree on the right. This example technically uses a more complex pruning method than the one described in the slides, but the general idea is the same.

Section 2

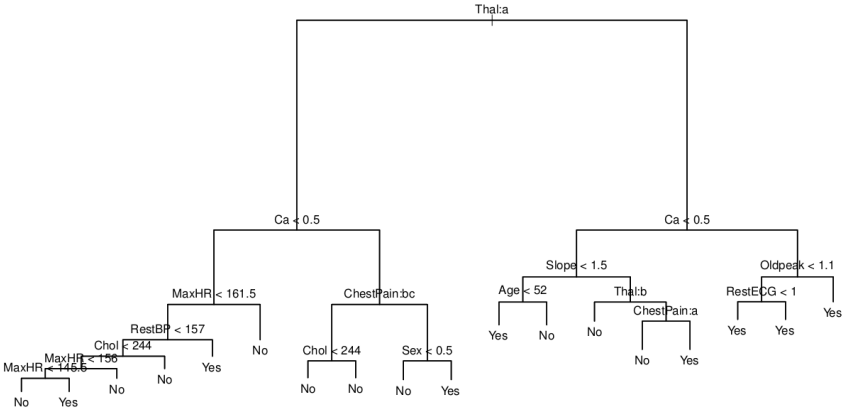
Classification Decision Tree

Basic idea

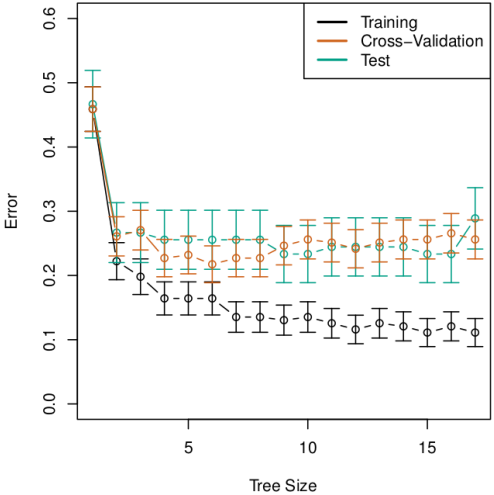


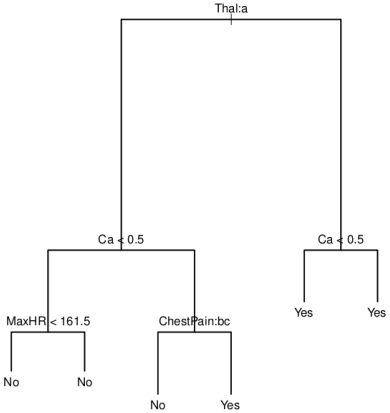
- $\hat{p}_{mk}$ = proportion of training observations in R_m from the kth class
- $E = 1 - \max_k(\hat{p}_{mk})$ *Fraction of training observations not in the most common class*
- Example from heart data
- Can't use RSS, need error rate
- Could decide splits by classification error rate
- Not necessarily the best, you can use other options but I'm going to skip that.

Example


Thal:a
Ca < 0.5
Ca < 0.5
MaxHR < 161.5
ChestPain:bc
RestBP < 157
Chol < 244
MaxHR < 156
MaxHR < 145.5
No
Yes
No
No
Yes
No
Yes
Chol < 244
Sex < 0.5
No
No
No
Yes
Slope < 1.5
Oldpeak < 1.1
Age < 52
Thal:b
RestECG < 1
Yes
Yes
Yes
ChestPain:a
Yes
No
No
No
Yes

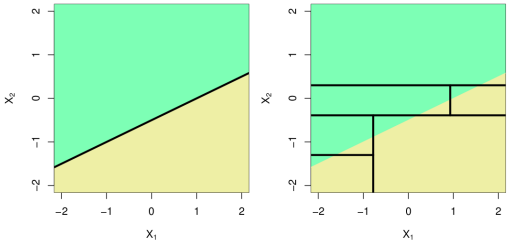
Pruning the example



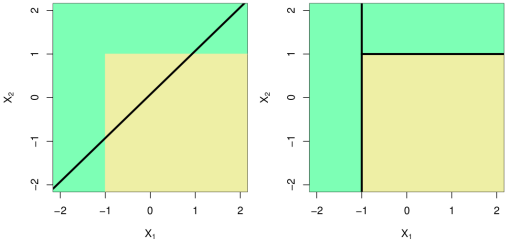


More coding!

Linear models vs trees



Obviously linear does better here



Not going to beat this case though

Pros/Cons

Pros:

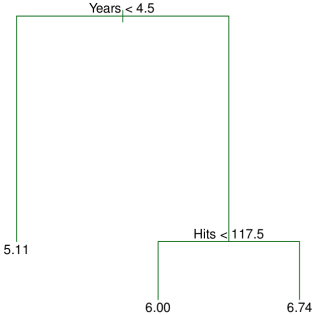
- Trees are very easy to explain to people. Often easier to explain than linear regression!
- Trees can be displayed graphically, and are easily interpreted even by a non-expert (especially if they are small).
- Trees can easily handle qualitative predictors without the need to create dummy variables.

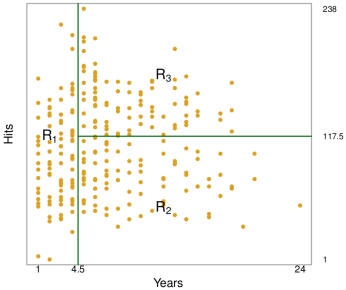
Cons:

- Not as accurate as other methods of classification and regression
- Not robust: small change in data can cause large change in estimated tree
- Fix..... aggregate many decision trees

TL:DR

- Split into regions by greedily decreasing RSS
- Prune tree by using cost complexity
- Not robust - Next time, figure out how to aggregate trees





Next time

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